

Software Engineering

MET CS-673 A1

Spring 2025

Instructor Name: Ronald Czik

Instructor Email: rec@bu.edu

Location and time: Wednesday – CAS B06B, 6:00 pm – 8:45 pm

Office hours: Mondays, 5:45 – 6:45 TBD

Assistant: Camille Christie, camchris@bu.edu Wednesday 4:30 – 5:30 by Zoom

Course Description

Overview of techniques and tools to develop high quality software. Topics include software development life cycle such as Agile and DevOps, requirements analysis, software design, programming techniques, refactoring, testing, AI in software engineering, as well as software management issues. This course features a semester-long group project where students will design and develop a real-world software system in groups using Agile methodology and various SE tools, including project management tools, programming frameworks, unit and system testing tools, integration tools and version control tools.

This is a capstone course to be taken after at least two programming intensive courses toward the end of a program of study. A solid understanding of Object-Oriented design concepts and proficiency in at least one high-level programming language is required. Familiarity with web or mobile application development preferred. This is a programming intensive course; everyone will be required to develop and debug code. No exceptions.

Books

- *Engineering Software Products - An Introduction to Modern Software Engineering* by Ian Sommerville – REQUIRED
- *Software Engineering* by Ian Sommerville; tenth edition - OPTIONAL
- Various other references as reviewed in class

Courseware

- All assignments and submissions will use Gradescope: <https://www.gradescope.com/courses/959499> Entry Code: DK7DEZ
- Discussions will be through Piazza: <https://piazza.com/bu/spring2025/cs673a1> Access Code: 1nem4dhet4t

Class Policies

- 1) **Attendance & Absences** – This is an experiential class; we will work in agile teams with a new scrum master each week. Course participation and collaboration with the instructor and peers is a requirement of the course. Attendance and participation are mandatory.
- 2) **Assignment Completion & Late Work** – Student performance will be evaluated on complete and timely submission of assignments and the results of quizzes and examinations and participation. Late homework assignments are not accepted. Do not wait until the last minute to begin an assignment. The lowest homework will be dropped. It is understood that students may need to miss a class due to unforeseen circumstances. To be fair to all students, the grading policy is to drop the lowest homework and to excuse one absence.
- 3) **Academic Conduct Code** – Cheating and plagiarism will not be tolerated in any Metropolitan College course. They will result in no credit for the assignment or examination and may lead to disciplinary actions. Please take the time to review the Student Academic Conduct Code:
http://www.bu.edu/met/metropolitan_college_people/student/resources/conduct/code.html.

This should not be understood as a discouragement for discussing the material or your particular approach to a problem with other students in the class. The purpose of assignments is to ensure students understand material presented in class. It is hoped that you will work with your peers on course assignments (and to prepare for exams). There is no penalty for working together. The goal is to ensure that everyone understands the material. If you choose to collaborate on course assignments, you are required to clearly disclose on the first page of your submission the individual(s) with whom you collaborated.

Grading Criteria

The grade that a student receives in this class will be based on class participation (including status reports and presentations), product artifacts (code, documentation, etc.), and individual work. The grade breakdown is shown below. All percentages are approximate, and the instructor reserves the right to make necessary changes.

Artifact	Weight
Product Proposal	10%
Individual assignments, ISRs, class participation	5%
PSRs	5%
Mid-semester product submission	20%
Final product submission	30%
Final exam	30%

Class Meetings, Lectures & Assignments

Lectures, Readings, and Assignments subject to change. Rev1, 1/19/2025

Class	Date	Topics	Readings	Assignments
1	1/22/2025	- Introduction – course outline - Software Products	Ch. 1	Project assigned
2	1/29/2025	Agile software engineering	Ch. 2	Assignment 1 due
3	2/5/2025	Features, Scenarios, and Stories	Ch. 3	- Project proposal due - Assignment 2 due - PRS, ISR, Meeting Minutes due
4	2/12/2025	Software Architecture	Ch. 4	
5	2/19/2025	Cloud-Based Software	Ch. 5	PRS, ISR, Meeting Minutes due
6	2/26/2025	Mid semester team presentations		Mid semester submission
7	3/5/2025	Microservices Architecture	Ch. 6	PRS, ISR, Meeting Minutes due
	3/12/2025	SPRING BREAK – NO CLASS		
8	3/19/2025	Security and Privacy	Ch. 7	
9	3/26/2025	Reliable Programming	Ch. 8	PRS, ISR, Meeting Minutes due
10	4/2/2025	Product review		
11	4/9/2025	Testing	Ch. 9	PRS, ISR, Meeting Minutes due
12	4/16/2025	DevOps and Code Management	Ch. 10	
	4/23/2025	SUBSTITUTE MONDAY SCHEDULE – NO CLASS		
13	4/30/2025	Team final presentations		Final project due
14	5/7/2025	Final exam		

